

Measuring How Prompt Compression Changes LLM Network Traffic

Preliminary local-model study on LoCoMo workloads

144

valid captures

2

application workloads

2

encrypted transports

Does reducing prompt tokens translate into fewer bytes on the wire?

We test the complete path:
conversation → compression

→ encrypted transport → LLM response

Measurements

- ▶ client-to-server bytes
- ▶ server-to-client bytes
- ▶ total captured bytes
- ▶ packets and latency

Two workloads expose different traffic regimes

Question answering

- ▶ 16 LoCoMo questions
- ▶ long conversation as input
- ▶ short factual answer
- ▶ output capped at 256 tokens

Input-dominated exchange

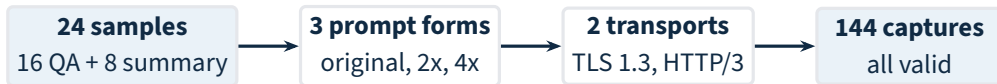
Event summarization

- ▶ 8 speaker-level tasks
- ▶ summarize one speaker's events
- ▶ long generated artifact
- ▶ output capped at 1,024 tokens

Output-dominated exchange

Source: LoCoMo benchmark: Maharana et al., ACL 2024.

Controlled experiment matrix



Model

Qwen3.5-9B via vLLM

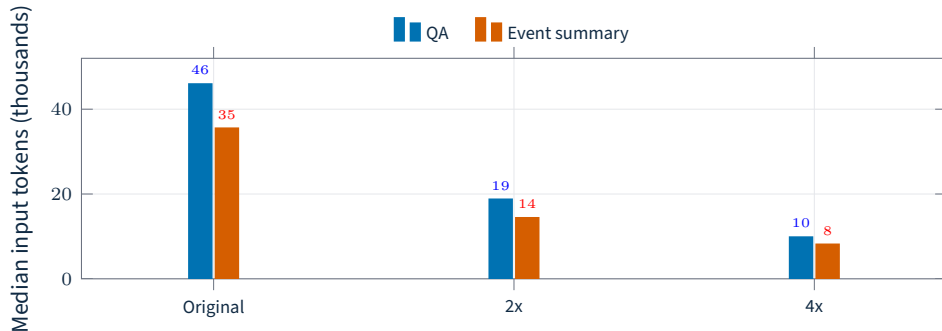
Determinism

temperature 0, seed 42

Network

loopback baseline

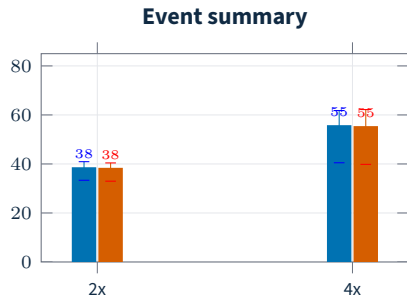
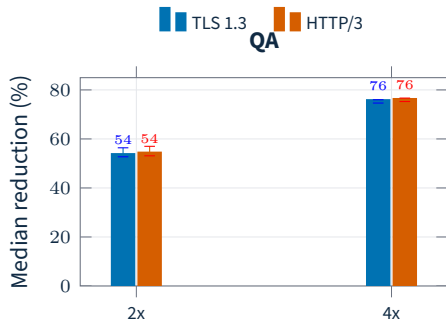
Compression reaches distinct token budgets



Median prompts shrink from 45.8k to 18.6k and 9.7k tokens for QA.

Source: This experiment; medians across 16 QA and 8 summary samples.

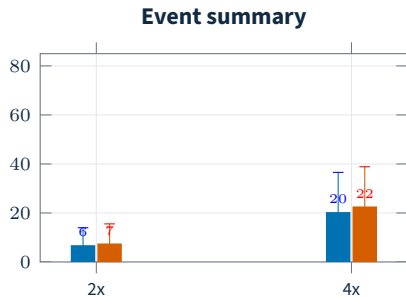
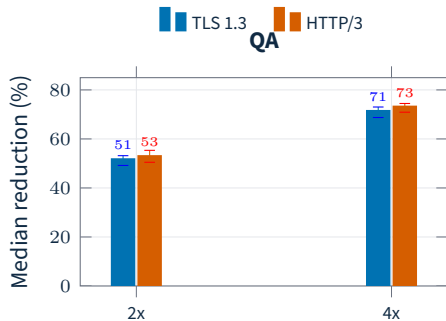
QA upload falls by 54–76%



The reduction is nearly identical under TLS 1.3 and HTTP/3.

Source: Paired medians; whiskers are deterministic bootstrap 95% confidence intervals.

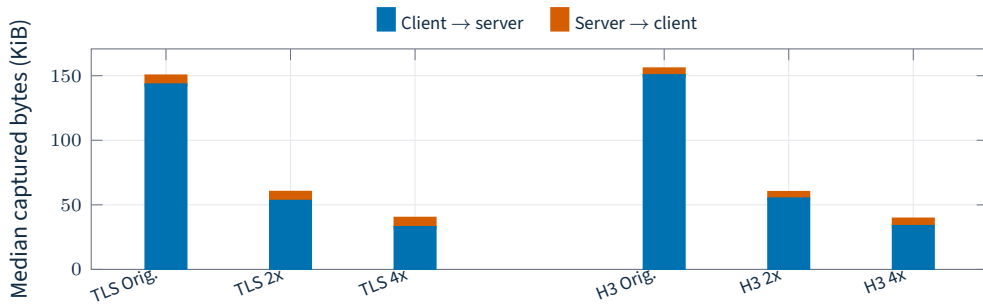
The workload determines total traffic savings



QA gains survive end-to-end; summary gains are diluted by large generated responses.

Source: Paired medians; bidirectional frame bytes.

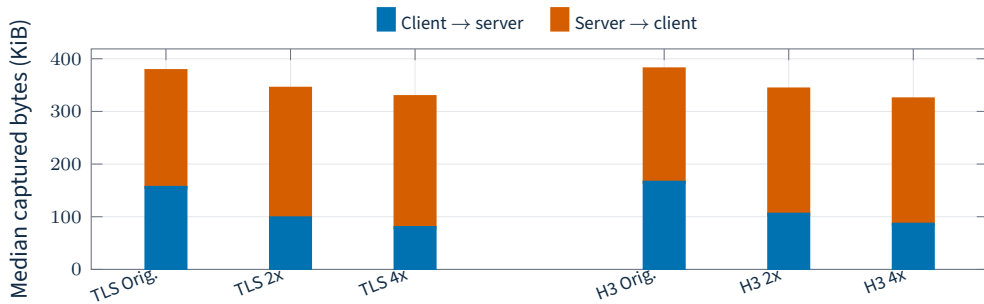
QA is dominated by the uploaded prompt



Compressing the input directly attacks the largest part of the exchange.

Source: Median captured bytes; one run per sample-condition pair.

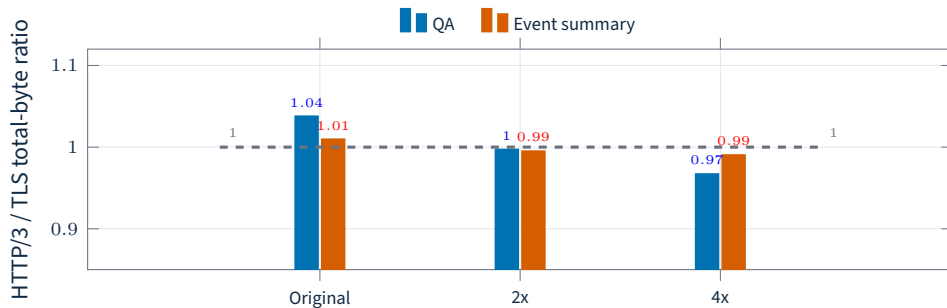
Generated summaries dominate bidirectional traffic



Upload shrinks, but response traffic remains roughly 220–255 KiB.

Source: Median captured bytes; one run per sample-condition pair.

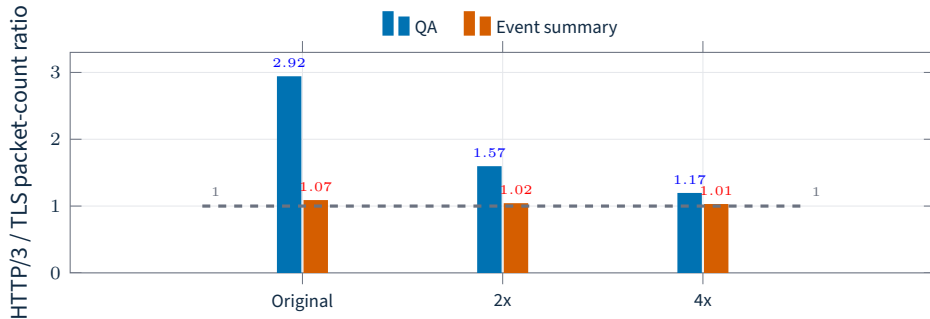
TLS and HTTP/3 carry similar total bytes



All median HTTP/3-to-TLS total-byte ratios remain within 4% of parity.

Source: Paired protocol ratios in a zero-RTT loopback baseline.

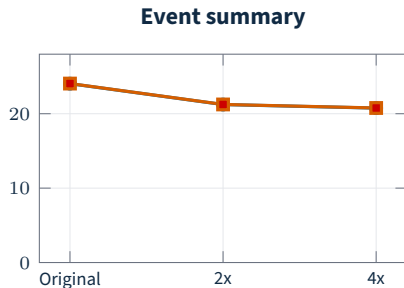
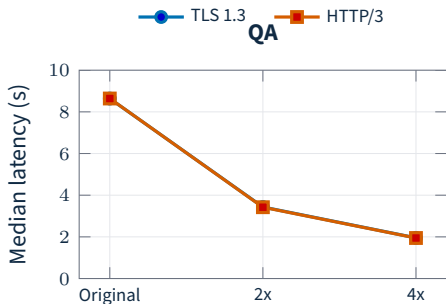
HTTP/3 packetization is visible on short exchanges



For original QA prompts, HTTP/3 uses 2.92x the captured packet count while transferring only 1.04x the bytes.

Source: Paired medians; packet count includes all captured protocol traffic.

Latency follows the amount of input processed



QA falls from 8.6 s to 2.0 s at 4x; summary falls from 24.0 s to 20.8 s.

Source: Median end-to-end generation latency; local model and loopback transport.

Three preliminary findings

- 01 Prompt tokens predict upload traffic.** Pearson $r = 0.88$ for both transports.
- 02 Compression value is workload-dependent.** At 4x, median total traffic falls 71–73% for QA but only 20–22% for summaries.
- 03 Protocol choice changes packetization more than bytes.** TLS and HTTP/3 are close in total traffic and latency on loopback.

Interpretation boundary and full-study plan

What these results establish

- ▶ end-to-end capture pipeline works
- ▶ compression levels are separable
- ▶ workload direction explains savings
- ▶ TLS and QUIC are directly comparable

Next controlled sweep

- ▶ 52 workload items
- ▶ loopback and realistic RTT
- ▶ three repetitions
- ▶ controlled MTU and offloads

$$52 \times 3 \times 2 \times 2 \times 3 = \mathbf{1,872 \text{ runs}}$$

**Prompt compression saves network traffic
when the prompt is the dominant payload.**

The next experiment tests whether this persists under realistic network conditions.

References and experiment artifacts

- ▶ Pan et al. *LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression*. arXiv:2403.12968, 2024.
- ▶ Maharana et al. *Evaluating Very Long-Term Conversational Memory of LLM Agents*. ACL 2024.
- ▶ Preliminary metrics: `artifacts/preliminary_metrics_baseline.csv`
- ▶ Reproducible figure pipeline: `analysis/prepare_figure_data.py`

All numerical claims in this deck are computed from the 144 valid baseline captures.